Linear regression – Mathematics

My model:

$$y = mx + c$$

Formula for the mean square error:

$$\mathbf{E} = \frac{1}{2n}\sum_{i=0}^{n}((mx_i + c) - y_i)^2$$

Finding the partial derivatives of the mean square error:

$$\frac{\partial E}{\partial m} = \frac{1}{n}\sum_{i=0}^{n}(x_i((mx_i + c) - y_i))$$

$$\frac{\partial E}{\partial c} = \frac{1}{n}\sum_{i=0}^{n}((mx_i + c) - y_i)$$

Formulas used for the gradient descent algorithm:

$$m = m - L(\frac{\partial E}{\partial m})$$

$$c = c - L(\frac{\partial E}{\partial c})$$